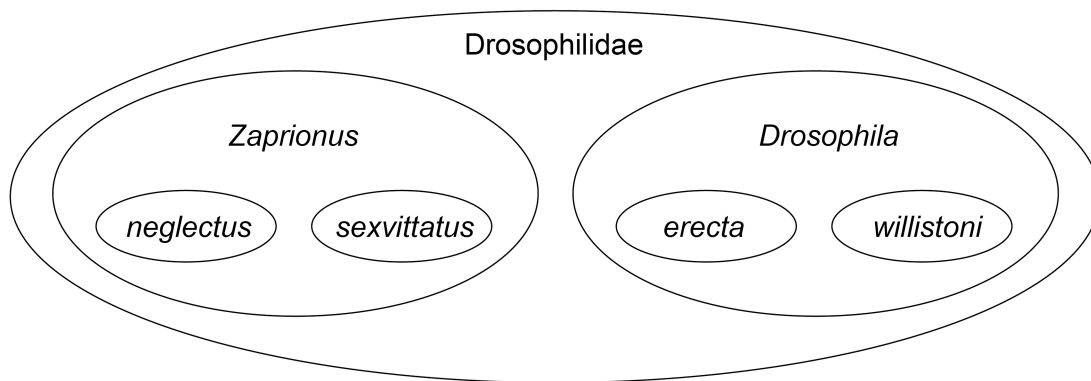


0 6

Figure 5 represents the phylogenetic classification of **four** different species of fruit fly.

Figure 5



0 6 . 1

Figure 5 shows a hierarchy. Explain how.

[2 marks]

0 6 . 2

Name the taxon to which Drosophilidae belongs.

[1 mark]

Turn over ►

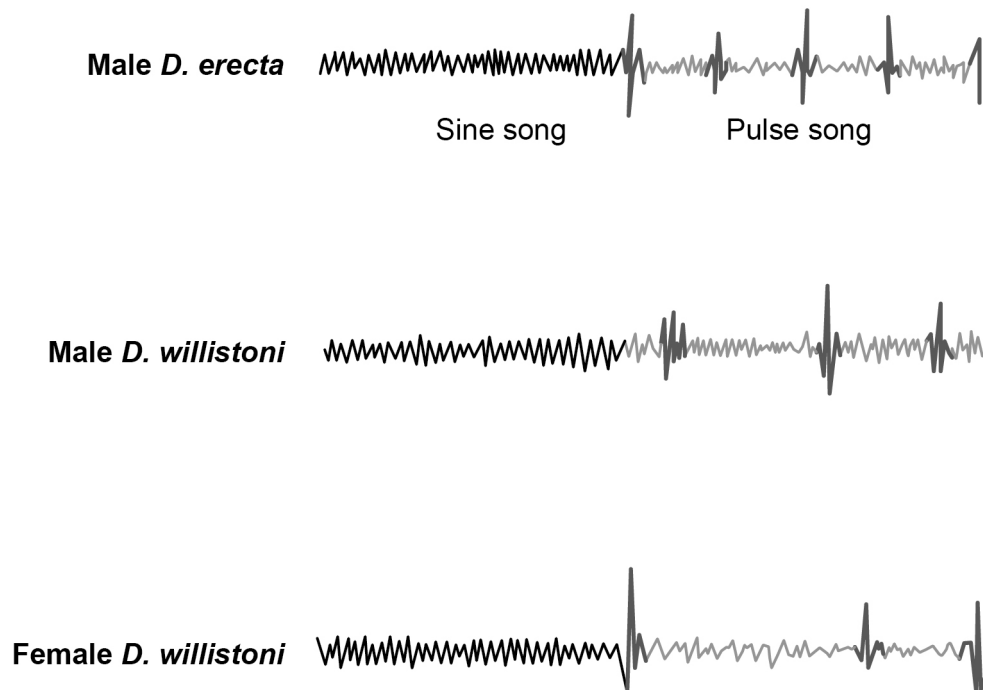


Drosophila fruit flies display courtship behaviour. One of the stages of courtship is singing by males. Normally a male will produce a 'sine song', in which continual noise is made, and a 'pulse song', in which there is continual noise with some louder peaks.

Scientists showed fruit flies a visual stimulus that made them sing. They made recordings of these songs.

Figure 6 shows the recordings of the songs of three flies over the same time period.

Figure 6



0 6 . 3 *D. erecta* and *D. willistoni* are closely related species **but** different species.

Describe evidence from **Figure 6** that supports this statement.

[2 marks]

0 6 . 4 The scientists repeated their experiments, using female fruit flies as the visual stimulus. When a male and female *D. willistoni* were together, their songs led to mating.

When two female *D. willistoni* were together, their songs did **not** lead to any attempt to mate.

Use information from **Figure 6** to suggest why the two females did not attempt to mate.

[2 marks]

